ISYE6240: Homework 2

Question 1

1a)

If $f(x) = \frac{3x^2}{16} \quad where - 2 < x < 2$ Then for evaluating $|X| < 0.5$:

$$P(-0.5 < X < 0.5) = \int_{-0.5}^{0.5} f(x)dx = \int_{-0.5}^{0.5} \frac{3x^2}{16}dx$$

$$P(-0.5 < X < 0.5) = \frac{x^3}{16}\Big|_{-0.5}^{0.5} = \frac{1}{128} - (-\frac{1}{128}) = \frac{1}{64}$$

1b)

The Cumulative Distribution Function is the integral of the PDF from it's lower limit to some point between the limits.

$$F(x) = \int_{-2}^{x} \frac{3t^2}{16}dt = \left[\frac{t^3}{16}\right]_{-2}^{x} = \frac{x^3}{16} - \frac{(-2)^3}{16} = \frac{x^3+8}{16}$$

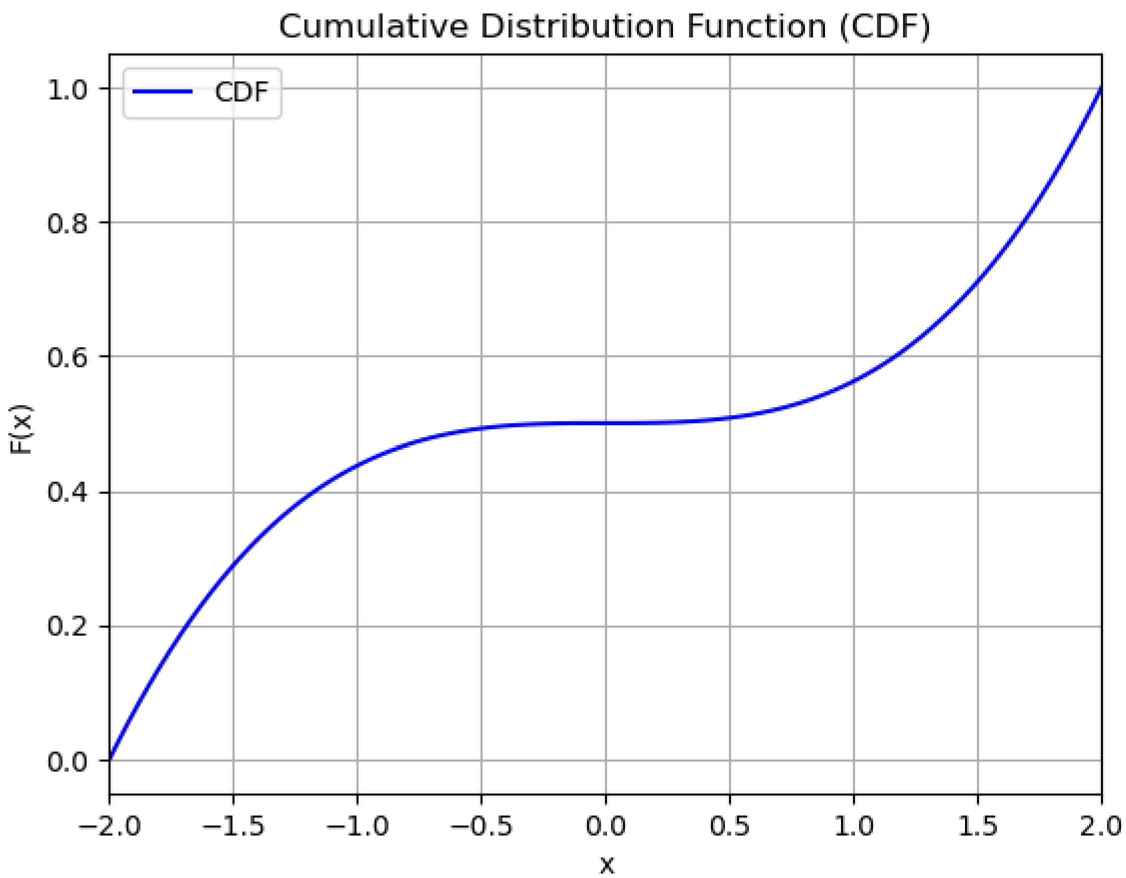
In [1]:

```
import matplotlib.pyplot as plt
# Plot CDF
limits = [-2, 2]
function = lambda x: ((x**3)+8)/16

def plot_cdf(func, x_limits, num_points=100):
    x = [x_limits[0] + i * (x_limits[1] - x_limits[0]) / (num_points - 1) for i in
    y = [func(xi) for xi in x]

    plt.plot(x, y, label='CDF', color='blue')
    plt.title('Cumulative Distribution Function (CDF)')
    plt.xlabel('x')
    plt.ylabel('F(x)')
    plt.xlim(x_limits)
    plt.grid(True)
    plt.legend()
    plt.show()

plot_cdf(function, limits)
```



1c)

The Expectation (the mean in this context) is calculated as

$$E(x) = \int x.\, f(x) dx$$

In the case where expectation $Y = X^2$ then:

$$EX = \int_{-2}^{2} x^2.\, f(x) dx = \int_{-2}^{2} \frac{3x^4}{16}$$

$$EX = \frac{3x^5}{80}\Big|_{-2}^{2} = \frac{96}{80} - (-\frac{96}{80}) = \frac{12}{5} = 2.4$$

Hence, the mean loss is $2400.

1d)

In the case of a loss being <4000, $Y = X^2$ = 4 so $X = \pm 2$

As these are the limits for out distribution, the maximum loss that can occur is $4000. Hence, the probability of loss being <4000 is 1.

Question 2

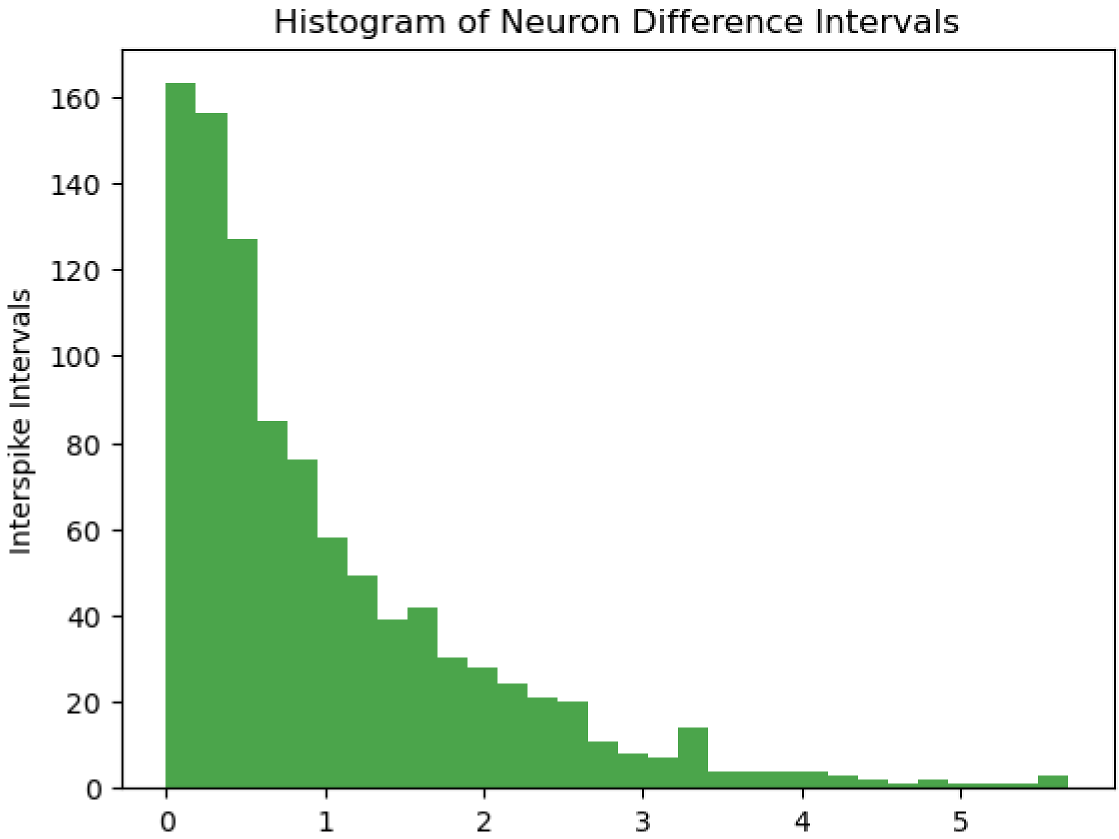
2a)

In [20]:
```
import pandas as pd
import matplotlib.pyplot as plt

# Import CSV data and plot histogram
data = pd.read_csv('../../data/neurondiffs.csv')
plt.hist(data, bins=30, color='green', alpha=0.7)
plt.title('Histogram of Neuron Difference Intervals')
plt.ylabel("Interspike Intervals")

# Calculate the MLE for the rate parameter of an exponential distribution
interspike_intervals = data['value']
mle_rate = 1 / interspike_intervals.mean()
print(f"MLE for the rate parameter (lambda) of the exponential distribution: {round
```

```
MLE for the rate parameter (lambda) of the exponential distribution: 0.99
```



The histogram of interspike intervals resembles an exponential distribution. Computing the MLE for the rate parameter is derived from finding the maximum of the likelihood function, which results in:

$$\hat{\lambda}_{MLE} = \frac{1}{\bar{T}} = \frac{n}{\sum T_i} = 0.99$$

where $\bar{T}$ is the sample mean.

Running this for the neurondiffs data produces and MLE of 0.99. For an exponential distribution if $\lambda \approx 1$, then the mean and standard deviation are approximately equal (with a value of approximately 1)

2b)

If the prior is $\lambda \sim \text{Gamma}(\alpha = 18, \beta = 20)$, and we know for our data (likelihood) we have the $\lambda_{data} = \lambda_{MLE} \approx 1$

We can utilise the kernel method, as in the conjugate family the posterior density is proportional to the likelihood and prior densities multiplied.

$$P(\lambda|T) \propto L(\lambda) \times \pi(\lambda)$$

The likelihood is the joint distribution of the PDFs.

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda T_i} = \lambda^n e^{-\lambda \sum T_i}$$

The Prior which is a Gamma distribution:

$$\pi(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda}$$

$$P(\lambda|T) \propto \lambda^{\alpha-1+n} e^{-\lambda(\beta+\sum T_i)}$$

Comparing with the intended form of the posterior distribution we can see the change from the prior as: $\alpha_{post} = \alpha + n$ and $\beta_{post} = \beta + \sum T_i$

We can find $\sum T_i$ as it's in the equation for $\lambda_{MLE} = \frac{n}{\sum T_i}$. Hence $\sum T_i = \frac{988}{0.99} \approx 997.98$

Therefore, the Bayes estimator (posterior mean) is:

$$E(\lambda|T) = \frac{\alpha + n}{\beta + \sum T_i} = \frac{18 + 988}{20 + 997.98} \approx 0.988$$

The posterior variance is:

$$Var(\lambda|T) = \frac{\alpha_{post}}{\beta_{post}^2} = \frac{\alpha + n}{(\beta + \sum T_i)^2} \approx \frac{1006}{1017.98^2} \approx 0.00097$$

2c)

If $\mu = \frac{1}{\lambda}$ and we want $P(\mu|T)$, then our new likelihood and prior are the following:

$$L(\mu) = \mu^{-n} e^{\frac{-\sum T_i}{\mu}}$$

$$\pi(\mu) \propto \mu^{-(a+1)} e^{\frac{b}{\mu}}$$

derived from the Inverse Gamma distribution of μ.

Using the Bayes Theorem for conjugate families as we did before: $P(\mu|T) \propto L(\mu)\pi(\mu)$

$$P(\mu|T) \propto \mu^{-(a+1)-n} e^{\frac{b}{\mu} - \frac{-\sum T_i}{\mu}} = \mu^{-(a+1+n)} e^{-\frac{1}{\mu}(b+\sum T_i)}$$

$$a_{post} = a + n,\ b_{post} = b + \sum T_i$$

We can now calculate posterior mean for posterior μ which also follows an inverse gamma distribution.

$$E(\mu|T) = \frac{b_{post}}{a_{post} - 1} = \frac{b + \sum T_i}{a + n - 1} = \frac{20 + 997.98}{18 + 988 - 1} \approx 1.01$$

Question 3

3a)

Marginal Posterior Distribution for a conjugate case:

$$P(\theta_1|y) \propto P(y|\theta_1). P(\theta_1)$$

The posterior for θ can be calculated as we know the Normal-Normal conjugate family above so our posterior is also a normal distribution

$$P(\theta \mid \text{data}) = \frac{P(\text{data} \mid \theta)P(\theta)}{P(\text{data})}$$

For conjugate family of normal-normal distributions:

$$P(\theta_1|y) \propto P(y|\theta_1). P(\theta_1)$$

$$\text{Posterior} \propto \exp\left(-\frac{(x-\theta)^2}{2\sigma^2} - \frac{(\theta-\mu)^2}{2\tau^2}\right)$$

The known formulas for calculating Posterior Mean and Posterior Variance are:

$$\mu_{post} = \frac{\tau^2}{\sigma^2 + \tau^2} y + \frac{\sigma^2}{\sigma^2 + \tau^2} \mu$$

$$\tau^2_{post} = \frac{\sigma^2 \tau^2}{\sigma^2 + \tau^2}$$

In this context:

$$Prior\ on\ \theta_i : N(\mu, \sigma) = N(0, 1)$$

$$Likelihood : N(\phi, \tau) = N(\theta_1 + \theta_2, 1)$$

Prior for $\theta_1 + \theta_2$ involves combining the two distributions. Combining two identical normal distributions is simply adding the means and variances, hence for $\phi = \theta_1 + \theta_2$:

$$Prior\ on\ \phi : N(\mu, \sigma) = N(0, 2)$$

This now allows us to use the above equations to calculate posterior mean and variance, when y=1:

$$\mu_{post} = \frac{2}{2+1} y + \frac{1}{2+1}(0) = \frac{2}{3} y = \frac{2}{3}$$

$$\tau^2_{post} = \frac{2 * 1}{2+1} = \frac{2}{3}$$

As θ_1 and θ_2 have identical iid priors and enter the likelihood symmetrically, their marginal posterior distributions are identical: $E[\theta_1 \mid y] = E[\theta_2 \mid y]$. Hence the posterior mean must be the sum of the individual means:

$$E[\theta_1 + \theta_2 \mid y] = E[\theta_1 \mid y] + E[\theta_2 \mid y] = 2/3$$

$$E[\theta_1 \mid y] = E[\theta_2 \mid y] = 1/3$$

Finding the marginal variance is more tricky, Since y is normally distributed with mean $\theta_1 + \theta_2$ and variance 1, we can represent it as a linear equation:

$$y = \theta_1 + \theta_2 + \epsilon$$

where $\epsilon \sim N(0, 1)$ represents the random noise and is independent to the other components.

We can use precision to calculate it where precision w for θ_1: $w = \frac{1}{\sigma^2}$ To find the posterior precision for θ_1, we sum the prior precision of θ_1 and the precision provided by the data (y).

$$w_{post} = w_{prior} + w_{data} = 1 + \frac{1}{1+1} = \frac{3}{2}$$

To find the posterior variance:

$$\tau^2_{post} = \frac{1}{w} = \frac{2}{3}$$

Finally:

$$\theta_1 | y \sim N(\frac{1}{3}, \frac{2}{3})$$

$$\theta_2 | y \sim N(\frac{1}{3}, \frac{2}{3})$$

3b)

For the posterior distribution of $\theta_1 - \theta_2$ we can first find the prior: As θ_1 and θ_2 are independent the combined mean and variance are simple to calculate, as we can remove the covariate term.

$$Var(a\theta_1 + b\theta_2) = a^2 Var(\theta_1) + b^2 Var(\theta_2)$$

If we say $D = \theta_1 - \theta_2$:

$$D \sim N(0, 1+1) \implies D \sim N(0, 2)$$

We know the likelihood is:

$$y|\theta_1, \theta_2 \sim N(\theta_1 + \theta_2, 1) \implies N(S, 1)$$

So the likelihood is only dependent on S = $\theta_1 + \theta_2$, where S is also independent of D as both are made up of independent components θ_1, θ_2 and as such we expect $Cov(\theta_1, \theta_2) = 0$. This can be proven with the covariance of linear combinations rule:

$$Cov(\theta_1 + \theta_2, \theta_1 - \theta_2) = Cov(\theta_1, \theta_1) + Cov(\theta_1, -\theta_2) + Cov(\theta_2, \theta_1) + Cov(\theta_2, -\theta_2)$$

$$Cov(S, D) = Var(\theta_1) - Cov(\theta_1, \theta_2) + Cov(\theta_2, \theta_1) - Var(\theta_2)$$

$$Cov(S, D) = 1 - 0 + 0 - 1 = 0$$

Thus we have shown that S and D are independent, so the likelihood is also independent of D. Hence the posterior of D is only dependent on the prior which we already know: